

전자소자개론(공) (EE302-01)

학기	2026년 2학기
담당교수	한상윤
이수구분	교선평수
학점	3.0
강의시간	월10:30-12:00(E7 - 224), 수10:30-12:00(E7 - 224)
강의실	E5 308 (교수실)
소속	기초학부

교과개요

For the last few decades, microelectronics industry has shown tremendous innovation and improvements. The ground-breaking innovations came along with the advancements of electronic devices. It enabled mobile electronic systems, and allowing analog/digital logic circuit implementations for various applications such as signal processing, optoelectronic devices, sensors, and photovoltaic cells. In this course, first, we will start by reviewing fundamentals of semiconductor physics. Then we will study electronic devices based on semiconductor materials in detail. After that, we will study modern semiconductor devices such as FinFET. At last, applications of the semiconductor devices in memory devices will be studied.

교과목표

- Understand operating principles of basic semiconductor device and there characteristics. - Learn applications of individual devices in system level.

주차별 강의계획

1주차:Semiconductor fundamentals 1
2주차:Semiconductor fundamentals 2
3주차:PN junction
4주차:Metal-semiconductor contacts (Schottky diodes)
5주차:Metal-oxide-semiconductor fundamentals 1
6주차:Metal-oxide-semiconductor fundamentals 2
7주차:MOSFETs: ideal
8주차:Mid-term exam
9주차:MOSFETs: electrostatics
10주차:MOSFETs: C-V characteristics
11주차:MOSFETs: nonideal
12주차:CMOS inverters
13주차:FinFET
14주차:Memory devices 1
15주차:Memory devices 2
16주차:Final exam

수업방법

강의(lecture)

학습윤리

No distribution of course materials without consent of the instructor. No cheating for all the course activities such as attendance, assignments, and exams.

교과정책(방침)

Attendance is highly recommended. Active participation in the lecture is encouraged.